

Global VTEC maps based on the ROB-IONO software and median polish kriging

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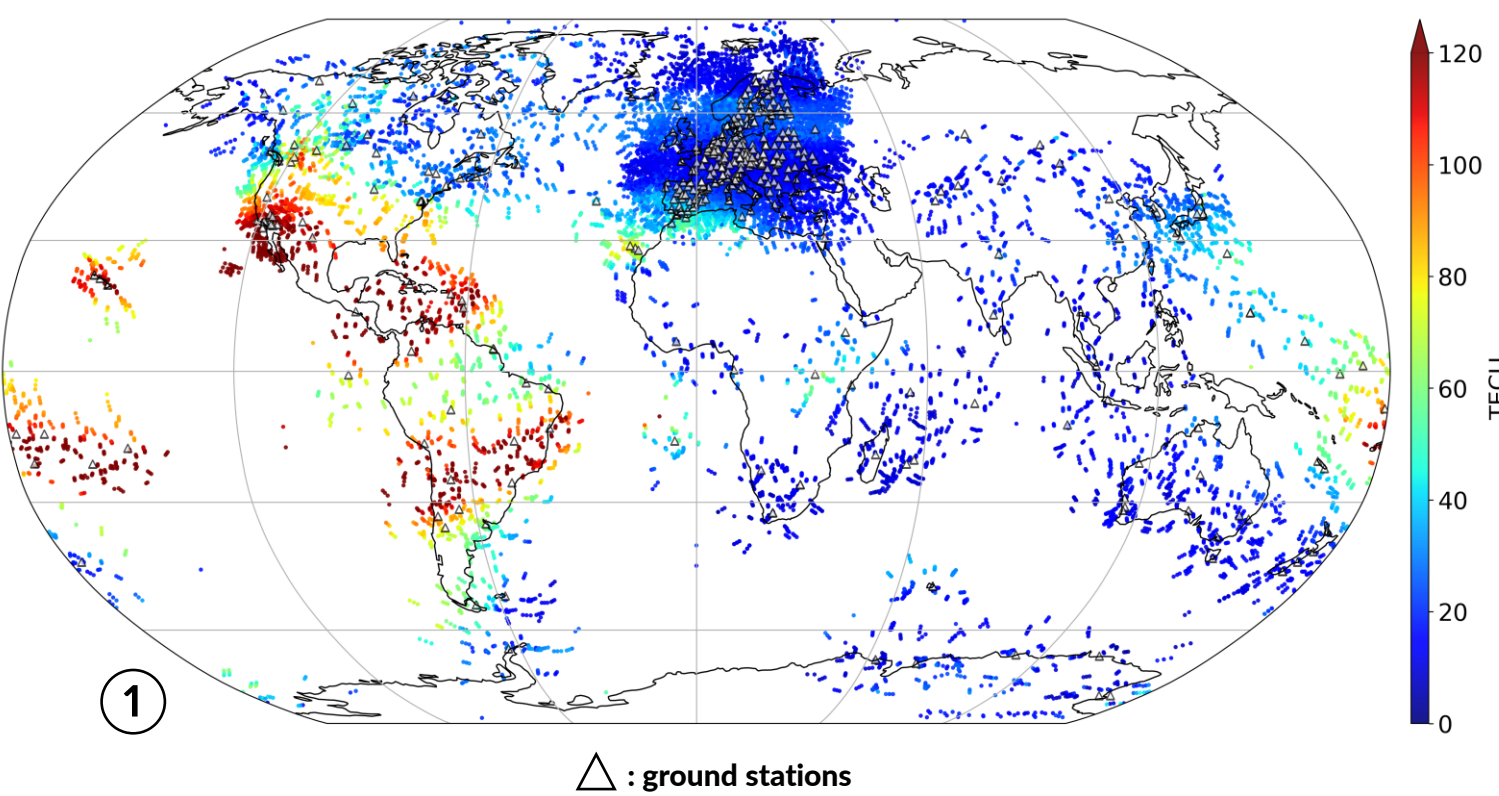


I. Methodology

1. Input Data

- VTEC at ionospheric piercing points (IPPs) ① from GNSS data and ROB-IONO software^[1], using IGS and EUREF station networks
- For example 15-minute interval:
~ 28 500 data points global,
~ 18 800 (66%) from EUREF

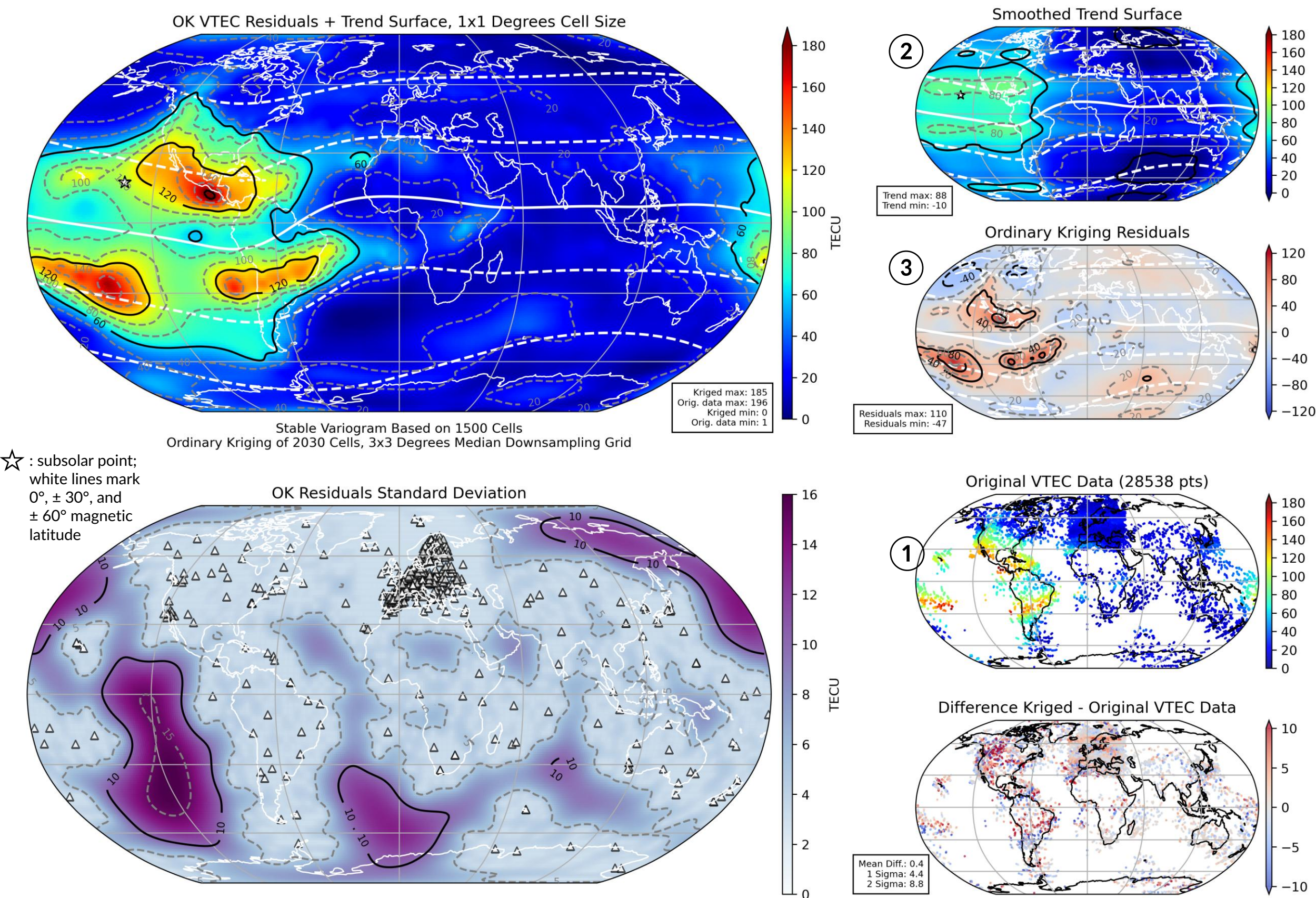
Example: 10 May 2024, DOY 131, 21:00–21:15 UTC



2. Median Polish and Ordinary Kriging (OK)

- Bin data into 3x3 degree cells, elevation-weighted median VTEC in each cell
- Iterative median polish^[e.g., 2, 3] with 30° longitude bins and 9° latitude bins to retrieve row and column (longitude and latitude) effects
- Compute smoothed (Savitzky-Golay filter) trend surface (1°x1° grid) ②
- Subtract trend surface from original data points to calculate new residuals
- Compute empirical variogram of residuals, then fit a stable function for ordinary kriging of residuals to a 1°x1° grid ③
- Add trend surface and kriged field for final map

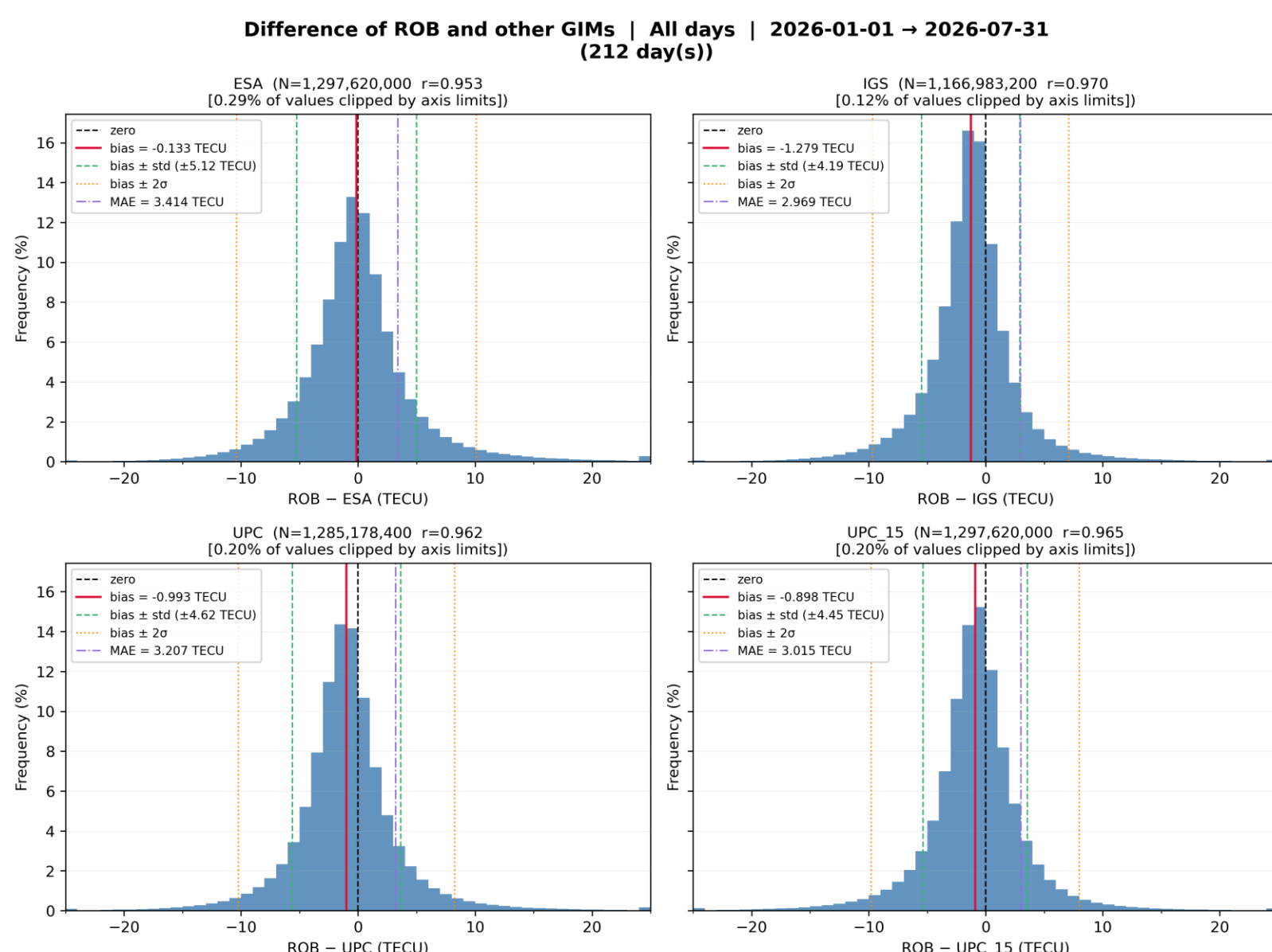
2024-05-10 | DOY 131 | 2100 + 15 min UTC



II. Comparison with other GIM products

The comparison with four other established GIM products shows low negative bias of our maps (-0.1 to -1.3 TECU) and no meaningful difference when excluding geomagnetically perturbed conditions.

Zonal difference maps show a more pronounced equatorial ionization anomaly (EIA) in our GIMs, which is even more visible during geomagnetically perturbed conditions.

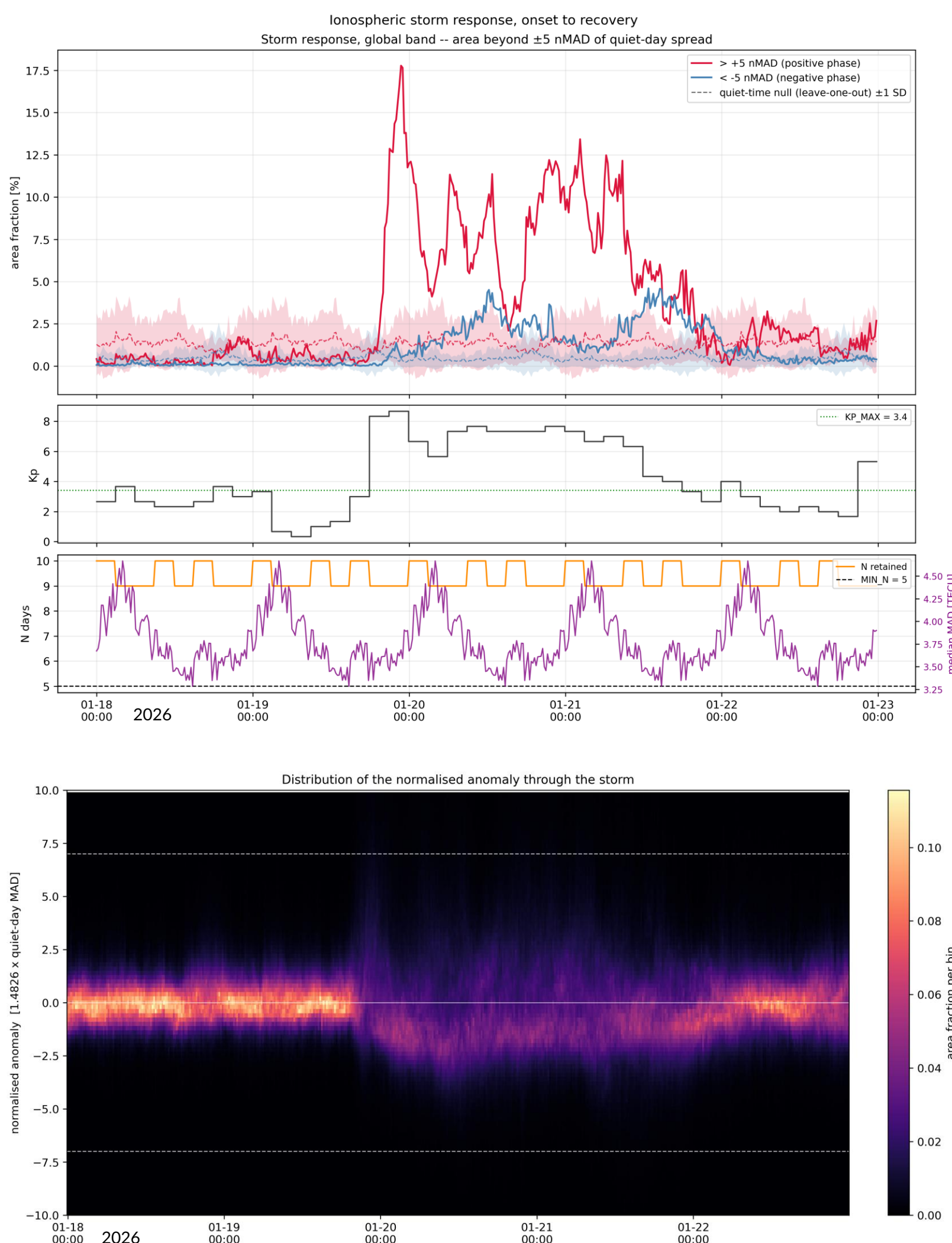


Motivation

- We aim to produce global maps of the vertical total electron content (VTEC) derived from dual-frequency GNSS observations recorded at IGS and EUREF ground stations.
- Entirely data-driven approach, requires no model input (except for determination of DCBs of receivers)
- Kriging interpolation allows for assessment of the precision of the result maps (variance/standard deviation)

III. Results and Scientific Application

- High-resolution global VTEC maps, with mean differences to the original data of ~0.2/0.4 TECU and standard deviations of differences of ~2/4 TECU (quiet/storm-time)
- Using a 14-day quiet baseline, excluding any perturbed epochs, we can establish the nMAD (normalised Median Absolute Deviation) as a robust reference of how much a cell's VTEC normally varies from one quiet day to the next. At each grid cell, the baseline is the median of the retained quiet maps.
- The 5x exceedance of nMAD as a global area fraction performs well as a global proxy for ionospheric disturbance, with both storm loading and recovery periods clearly visible in the data (red and blue lines below, respectively).



The ROB GIMs are currently produced with a 1-day lag and we aim to make them available on the website (gnss.be) next to the near-real time European maps in the future.

The corresponding paper is in preparation and will be submitted in October 2026.

References

- Bergeot et al. (2014), DOI: 10.1051/swsc/2014028
- Cressie (1986), DOI: 10.2307/2288990
- Cressie (1993), DOI: 10.1002/9781119115151

